

RESULTS

Results

01 · GRANGER CAUSALITY

does X predict future Y?

Table. Granger tests (both directions)

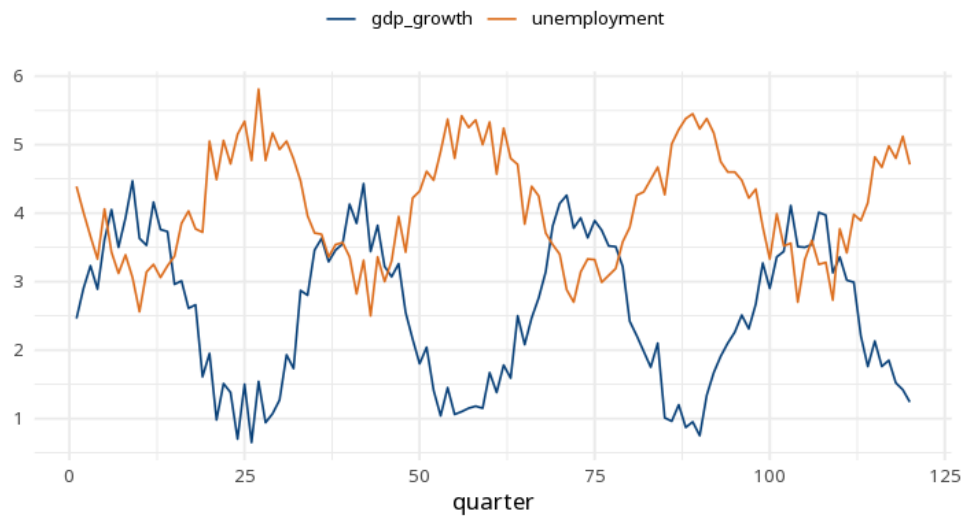
Direction	F	df	p
X→Y	122.15	4, 111	<.001
Y→X	3.42	4, 111	.011

Table 2. Lag-order selection (advisory)

Lag	AIC	BIC	HQ
1	-5.14	-5.00	-5.08
2	-5.14	-4.90	-5.04
3	-5.17	-4.84	-5.04
4	-5.23	-4.81	-5.06

tests predictive precedence, not true causation; both series should be stationary first. Tests use a max lag of 4; select the lag on theory or by AIC/BIC (vars::VARselect) and report it.

Figure. Series together



Understanding the numbers

Direction. Which direction this row tests - whether the first series helps predict the second, or vice versa. Here, this row tests $X \rightarrow Y$.

F. The F-statistic testing whether past values of one series improve prediction of the other, beyond the other's own past. Here, $F = 122.15$.

df. The degrees of freedom for the F-test (numerator, denominator). Here, $df = (4, 111)$.

p. A significant p means past values of X help predict Y beyond Y's own past - predictive precedence, not proof of causation. Here, $p < .001$.

Lag. A candidate lag order compared by information criteria - the note above uses a fixed max lag regardless of what this table recommends. Here, AIC is minimized at lag = 4.

AIC. An information criterion used to pick the lag order - lower is better relative to the alternatives compared. Here, AIC = -5.23 at the AIC-minimizing lag.

BIC. Like AIC but penalizes complexity more heavily; used alongside it to pick the lag order. Here, BIC = -4.81 at the AIC-minimizing lag.

HQ. The Hannan-Quinn criterion: another information criterion for comparing candidate lag orders. Here, HQ = -5.06 at the AIC-minimizing lag.

How to read this test

A significant $X \rightarrow Y$ p means past values of X help predict Y beyond Y's own past - this is predictive precedence, not proof that X causes Y. Check both directions. The result depends entirely on the lag order (this card uses a max lag of 4): choose it on theory or by an information criterion - `vars::VARselect` reports the AIC- and BIC-minimising lag - and report it, since different lags can flip the conclusion. Make both series stationary first. Method: `R lmtest::grangertest` (Granger, 1969).

Your significance threshold (α) is 0.05.

APA template: Granger test $X \rightarrow Y$: $F(4,111)=122.15$, $p < .001$; $Y \rightarrow X$: $F(4,111)=3.42$, $p = .011$ (lag=4).

Statistical basis: Granger causality F-test (Granger, C. W. J. (1969). "Investigating causal relations by econometric models and cross-spectral methods." *Econometrica*, 37(3), 424-438.); Implementation (`lmtest::grangertest`) (Zeileis A, Hothorn T (2002). "Diagnostic Checking in Regression Relationships." *R News*, 2(3), 7-10.). Full references in the exported CITATIONS.txt.

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